

CHRISTINA DURÓN

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🌐 <https://cduron.info>

ACADEMIC EMPLOYMENT

Assistant Professor of Mathematics

Natural Science Division of Seaver College, Pepperdine University

Aug 2022 – Present

Postdoctoral Research Associate

Department of Mathematics, University of Arizona

Aug 2019 – May 2022

High School Teacher

Mathematics Department, The Webb Schools of California

Aug 2013 – June 2019

RESEARCH INTERESTS

Network Theory; Network Dynamics; Statistical Analysis and Modeling of Complex Networks; Mathematical-Biology

EDUCATION

Claremont Graduate University

Ph.D. in Mathematics

May 2019

- **Thesis:** The Distribution of Betweenness Centrality in Exponential Random Graph Models
- **Advisors:** Dr. Ami Radunskaya (Professor, Pomona College) and Dr. Johanna Hardin (Professor, Pomona College)

University of Washington

Master in Science in Applied Mathematics

June 2013

Swarthmore College

Bachelor of Arts in Mathematics; Minor in Computer Science

May 2012

PUBLICATIONS *

13. Kravitz H, **Durón C**, Nieves B, Brio M. (2025). Data-driven optimization and parameter estimation for a metric graph epidemic model with applications to COVID-19 spread in Poland: A real-world example of optimization for a challenging Rosenbrock-type objective function. *An International Journal of Optimization and Control: Theories & Applications (IJOCTA)*, 15(4). doi: [10.36922/IJOCTA025220106](https://doi.org/10.36922/IJOCTA025220106)
12. **Durón C**, Kravitz H, Brio M. (2025). Edge percolation centrality: A new measure to quantify the influence of edges during percolation in networks. *PLoS One* 20(9): e0331475. doi: [10.1371/journal.pone.0331475](https://doi.org/10.1371/journal.pone.0331475)
11. **Durón C**, Kravitz H, Brio M. (2025). Kolmogorov–Smirnov-Based Edge Centrality Measure for Metric Graphs. *Dynamics*, 5(2), 16. doi: [10.3390/dynamics5020016](https://doi.org/10.3390/dynamics5020016)
10. Pfeffer D, **Durón C**. (2025). An Unlikely Duo: Injecting Art Projects in the Mathematics Classroom. *Journal of Humanistic Mathematics*, 15(1), 87-108. doi: [10.5642/jhummath.RDDV3141](https://doi.org/10.5642/jhummath.RDDV3141)
9. Kravitz H, **Durón C**, Brio M. (2024). A Coupled Spatial-Network Model: A Mathematical Framework for Applications in Epidemiology. *Bulletin of Mathematical Biology*, 86(132): 1 – 35. doi: [10.1007/s11538-024-01364-3](https://doi.org/10.1007/s11538-024-01364-3)

* Authors are ordered by contribution.

† Undergraduate student.

8. **Durón C**, Swansen B[†], Kravitz H. (2024). Difference Approximation. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: [10.1002/9781118445112.stat08440](https://doi.org/10.1002/9781118445112.stat08440)
7. Sullivan N, **Durón C**, Pfeffer D. (2023). From Mirrors to Wallpapers: A Virtual Math Circle Module on Symmetry. *Journal of Math Circles*: Vol. 3: Iss. 1, Article 1. Available at <https://digitalcommons.cwu.edu/mathcirclesjournal/vol3/iss1/1>.
6. **Durón C**. (2022). Adaptive Quadrature. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: [10.1002/9781118445112.stat08388](https://doi.org/10.1002/9781118445112.stat08388)
5. **Durón C**, Farrell A. (2022). A Mean-Field Approximation of SIR Epidemics on an Erdős-Rényi Network Model. *Bulletin of Mathematical Biology*, 84(7): 1 – 19. doi: [10.1007/s11538-022-01026-2](https://doi.org/10.1007/s11538-022-01026-2)
4. **Durón C**. (2021). Linear Algebra, Computational. *Wiley StatsRef: Statistics Reference Online* (eds N. Balakrishnan, T. Colton, B. Everitt, W. Piegorsch, F. Ruggeri and J.L. Teugels). doi: [10.1002/9781118445112.stat00459.pub2](https://doi.org/10.1002/9781118445112.stat00459.pub2)
3. **Durón C**. (2020). Heatmap Centrality: A New Measure to Identify Super-Spreader Nodes in Scale-Free Networks. *PLoS ONE*, 15(7): e0235690. doi: [10.1371/journal.pone.0235690](https://doi.org/10.1371/journal.pone.0235690)
2. **Durón C**, Pan Y, Gutmann D.H., Hardin J, & Radunskaya A. (2018). Variability of Betweenness Centrality and Its Effect on Identifying Essential Genes. *Bulletin of Mathematical Biology*, 81(9): 3655 – 3673. doi: [10.1007/s11538-018-0526-z](https://doi.org/10.1007/s11538-018-0526-z)
1. Pan Y, **Durón C**, Bush E.C., et al. (2018). Graph Complexity Analysis Identifies an ETV5 Tumor-Specific Network in Human and Murine Low-Grade Glioma. *PLoS ONE*, 13(5): e0190001. doi: [10.1371/journal.pone.0190001](https://doi.org/10.1371/journal.pone.0190001)

UNDER REVISION *

2. O'Brien E, **Durón C**. The Wasserstein metric as a tool to assess burn-in and convergence. *Statistical Science*.
1. **Durón C**, Mello J[†], Kamps K[†]. Network Evaluation of Influential Sensors on Critical Infrastructures. *The Journal of Dam Safety*.

IN PREPARATION *

1. **Durón C**, Parkinson C, Kravitz H, Brio M, Amlicke S[†], DiLellio R[†], Wexelman R[†]. Modeling Epidemics with Noncompliance: A Discrete-Time Mean-Field Approach. *Bulletin of Mathematical Biology*.

RESEARCH POSITIONS

Graduate Research Assistant

Pomona College

Jan 2017 – June 2018

- NIH (1R01-CA195692-01) funding under Dr. Ami Radunskaya and Dr. Johanna Hardin

Jet Propulsion Laboratory (JPL) Intern

California Institute of Technology

June 2015 – August 2015

- Implemented the Extended Kalman Filter (EFK) and incorporated inter-robot measurements to improve the state estimation and localization of autonomous vehicles

Mathematical and Theoretical Biology Institute Researcher

Arizona State University

June 2011 – July 2011

- Developed a mathematical model for the evaluation and analysis of the air pollution in Los Angeles

* Authors are ordered by contribution.

† Undergraduate student.

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† Undergraduate student.

TEACHING EXPERIENCE

Instructor of Record

Pepperdine University

- **MATH 450: Mathematical Statistics** *Spring 2024*
- **MATH 350: Mathematical Probability** *Fall 2025, Fall 2023*
- **MATH 345: Numerical Methods** *Spring 2026*
- **MATH 260: Linear Algebra** *Spring 2026, Fall 2024, Spring 2023*
- **MATH 150: Calculus I** *Fall 2025, Fall 2024, Spring 2024, Fall 2023, Spring 2023, Fall 2022*

Instructor of Record

University of Arizona

- **MATH 491: Undergraduate Teaching Assistantship (UTA) Seminar** *Fall 2021 – Spring 2022*
- **MATH 196M: Calculus I Supplementary Seminar** *Spring 2022*
- **MATH 396L: Wildcats Proofs Workshop** *Spring 2022*
- **MATH 464: Theory of Probability** *Fall 2021*
- **MATH 363: Introduction to Statistical Methods** *Spring 2021*
- **MATH 129: Calculus II** *Fall 2020*
- **MATH 475A: Mathematical Principles of Numerical Analysis** *Fall 2020*
- **MATH 163: Basic Statistics** *Spring 2020*
- **MATH 122B: First Semester Calculus** *Fall 2019*
- **MATH 196L: Precalculus Supplementary Seminar** *Fall 2019*

Instructor of Record

The Webb Schools of California

- **Advanced Placement Computer Science Principles** *Fall 2018 – Spring 2019*
- **Honors Precalculus** *Fall 2014 – Spring 2019*
- **Introduction to Computer Programming with Python** *Fall 2014 – Spring 2018*
- **Precalculus** *Fall 2013 – Spring 2019*
- **Integrated Mathematics 2** *Fall 2013 – Spring 2014*

PRESENTATIONS

Student-Directed

4. **Modeling Epidemics with Noncompliance: A Discrete-Time Mean-Field Approach** *April 2026*
Pepperdine University; Seaver College Research and Scholarly Research Symposium
3. **Network Evaluation of Influential Sensors: A Proposed Approach** *March 2024*
Pepperdine University; Seaver College Research and Scholarly Research Symposium

2. **Using Networks to Evaluate an Infrastructure's Condition During Seismic Activity: A Proposed Approach** Jan 2024
Joint Mathematics Meeting: AMS - PME Undergraduate Student Poster Session
1. **Using Networks to Evaluate an Infrastructure's Condition During Seismic Activity: A Proposed Approach** July 2023
Pepperdine University, Natural Science Division Summer Research

Contributed

9. **Edge Percolation Centrality: Measuring Dynamic Influence on Networks** March 2026
American Mathematical Society Special Session on Metric Graphs: Theory and Applications
8. **Digitally-Enabled, Evidence Based Homework: Developing a Custom Platform to Improve Calculus Learning** January 2026
Joint Mathematics Meeting Special Session on Digitally-Enabled Evidence-Based Teaching Practices
7. **A Coupled Spatial-Network Model for Epidemiology** July 2024
Frankfurt School of Finance and Management Workshop on Metric Networks
6. **Math Circles: Elevating Student Engagement and Mathematical Enjoyment Outside the Classroom** Jan 2024
University of Arizona Mathematics Educator Appreciation Day Conference (MEAD)
5. **Incorporating Undergraduate Students in Research on Assessing Dam Safety Using Network Analysis** Jan 2024
Joint Mathematics Meeting Special Session on Navigating the Benefits and Challenges of Mentoring Students in Data-Driven Undergraduate Research Projects
4. **Tiling a Chessboard: A Problem Adapted for the Virtual Math Circle** Jan 2023
Joint Mathematics Meeting Special Session on Math Circle Activities as a Gateway into Mathematics
3. **A Mean Field Approximation of SIR Epidemics on an Erdős-Rényi Network Model** May 2021
Los Alamos-Arizona Days Conference (Virtual Poster)
2. **Identifying Super-Spreader Nodes in Scale-Free Networks using Network Centrality Measures** Sept 2020
Arizona Postdoctoral Research Conference (Virtual Talk)
1. **Identifying Treatment Targets for Pediatric Gliomas using Network Centrality Measures** June 2020
SIAM Conference on the Life Sciences (Virtual Talk)

Seminar

4. **A Coupled Spatial-Network Model for Epidemiology** Nov 2023
University of Florida, Health Systems Medicine Seminar (Virtual Talk)
3. **Network Centrality: Theory to Applications** Oct 2021
Arizona State University, Mathematical Biology Seminar (Virtual Talk)
2. **Heatmap Centrality: A New Measure to Identify Super-Spreader Nodes in Scale-Free Networks** Feb 2021
Claremont Colleges and University of Utah, Joint Applied Mathematics Seminar (Virtual Talk)
1. **Network Data Analysis Techniques on DESeq and RNASeq Data** Nov 2019
University of Arizona, TRIPODS Research Working Group 6 - Analyzing large-scale point-set data

Industry

1. **Network Evaluation of Influential Sensors: A Proposed Approach[‡]** March 2024
California Department of Water Resources
2. **Network Evaluation of Influential Sensors: A Proposed Approach[‡]** Nov 2023
Southern California Edison, Dam & Public Safety Group

[‡] Presented with undergraduate students.

[‡] Presented with undergraduate students.

PROFESSIONAL DEVELOPMENT

Program

- **American Institute of Mathematics SQuaRE Program** *April 2025*
California Institute of Technology
 - Member of a multi-year research collaboration focused on creating computational frameworks to predict epidemic transitions and optimize social measures for disease mitigation.
- **Project NExT** (Green 2023) *Aug 2023 – Aug 2024*
Mathematical Association of America (MAA)
 - Focused on all aspects of an academic career: improving the teaching and learning of mathematics, engaging in research and scholarship, finding exciting and interesting service opportunities, and participating in professional activities

Certification

- **Diversity, Equity, and Inclusion in the Workplace** *May 2021*
University of South Florida
 - Focused on ways that organizations can create a more diverse workplace, address equity issues, and foster inclusivity
- **Effective Online Discussions** *June 2020*
University of Arizona
 - Developed strategies for designing and facilitating effective online discussions that deepen learning, expand student exposure to curriculum, and increase student engagement
- **Teaching the Large Online Course** *June 2020*
University of Arizona
 - Developed instructional practices for encouraging student engagement and motivation in a large online class, as well as for effectively managing administrative tasks such as monitoring student progress and conducting assessments

Workshops

- **Workshop on Metric Networks** *July 2024*
Frankfurt School of Finance and Management
- **Broadening Participation: 2024 Mathematical and Physical Sciences Workshop for New Investigators** *July 2024*
National Science Foundation
- **Network Modeling for Epidemics** *Aug 2020*
University of Washington
- **BioBridge Clinic** *Jan 2020*
University of California, Irvine
- **Computational Genomics Summer Institute** *May 2019*
University of California, Los Angeles

SERVICE

Mentoring and Advising

- **Undergraduate Research Advisor** (Team of 3 undergraduates) *Fall 2025 – Spring 2026*
Pepperdine University
- **Tidy Thursday Advisor** *Spring 2026, Fall 2025*
Pepperdine University
- **Undergraduate Research Advisor** (Team of 2 undergraduates) *Summer 2023 – Fall 2024*
Pepperdine University

- **Math Club Vice-Principal Faculty Advisor** *Pepperdine University* *Spring 2023 – Fall 2023*
- **Undergraduate Student Mentor** *Pepperdine University, Faculty/Student Mentor Program* *Fall 2025, Spring 2023 – Fall 2023*
- **Undergraduate Student Mentor** *University of Arizona, Women in Science and Engineering (WISE) Program* *Fall 2021 – Spring 2022*
- **Graduate Student Mentor** *University of Arizona, Association for Women in Mathematics (AWM) Mentor Network* *Fall 2021 – Spring 2022*
- **Undergraduate Research Advisor** *University of Arizona* *Fall 2020 – Spring 2021*
- **Mathematics Undergraduate Teaching Assistantship (UTA) Program Mentor** *University of Arizona* *Fall 2020 – Spring 2021*
- **Mathematical Modeling Group Mentor** (Team of 4 undergraduates) *University of Arizona* *Spring 2020*
- **Math Club Advisor** *The Webb Schools of California* *Fall 2017 – Spring 2019*

University and Departmental Service

- **Creator and principal faculty advisor for Tidy Thursday** *Pepperdine University* *Spring 2026, Fall 2024*
 - An unofficial student club focused on exploring and analyzing datasets in **R** to improve data analysis and visualization skills.
- **Institutional Review Board, Natural Science Representative** *Pepperdine University* *Fall 2023 – Present*
 - Review research study protocol/application materials (exempt, expedited, and full review) and evaluate them from the perspective of the regulatory criteria for approval.
- **Admissions and Scholarship Committee** *Pepperdine University* *Fall 2025 – Present, Fall 2023 – Fall 2024*
 - Review student applications and select Faculty-Staff Scholars based on merit, including academic performance and recommendations.
- **Co-Instructor of RISE/Connection/Welcome Chapel** *Pepperdine University* *Spring 2026, Fall 2025, Fall 2023 – Fall 2024*
 - Facilitated discussions on different aspects of science from the Christian perspective, primarily in the disciplines of history, physics, astronomy, and mathematics, with students over five sessions
- **Member of Hiring Committee, Assistant Professor of Mathematics (AY 24/25)** *Pepperdine University* *Fall 2023 – Spring 2024*
 - Served on the seven-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom and in-person interviews on campus, collecting and consolidating information about the interviews, and making final recommendations
- **Member of Hiring Committee, Visiting Assistant Professor of Mathematics (AY 23/24)** *Pepperdine University* *Summer 2023*
 - Served on the five-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom, collecting and consolidating information about the interviews, and making final recommendations
- **Member of Hiring Committee, Assistant Professor of Mathematics (AY 23/24)** *Pepperdine University* *Fall 2022 – Spring 2023*

- Served on the seven-member review committee whose duties included reading through application packages, conducting virtual interviews via Zoom and in-person interviews on campus, collecting and consolidating information about the interviews, and making final recommendations
- Member of Review Committee, Excellence in Postdoctoral Mentoring Award** *Spring 2022*
University of Arizona
 - Served on the three-member review committee that determined the recipient of the 2022 Excellence in Postdoctoral Mentoring Award
- President, Postdoctoral Governance** *Fall 2021 – Spring 2022*
University of Arizona
 - Serve as an in-between for the postdocs and the Postdoctoral Committee, and organize the postdoctoral professional development seminar topics and panels
- Mathematics Undergraduate Teaching Assistantship (UTA) Program, Director** *Fall 2021 – Spring 2022*
University of Arizona
 - Coordinate the mentorship of the UTA's, and run the weekly professional development seminar
- Mathematics Undergraduate Teaching Assistantship (UTA) Program, Co-Director** *Fall 2020 – Spring 2021*
University of Arizona
 - Supported the Director of the UTA Program, and was responsible for additional duties related to the weekly professional development seminar
- Vice President, Postdoctoral Governance** *Fall 2020 – Spring 2021*
University of Arizona
 - Supported the President of the Postdoctoral Governance, and was responsible for additional duties related to the postdoctoral professional development seminars
- Non-Academic Liaison, Postdoctoral Governance** *Spring 2020*
University of Arizona
 - Organized a panel pertaining to non-academic careers for the postdoctoral professional development seminar

Service to the Discipline

- American Mathematical Society Spring Western Sectional Meeting 2026, Co-Organizer of Special Session** *March 2026*
Boise State University
 - “Metric Graphs: Theory and Applications”
- Webinar Panelist** *Feb 2026*
Transform Learning
 - “Thinking About Thinking: Using Formative Practice to Grow Metacognitive Learners”
- Joint Mathematics Meeting 2024, Co-Organizer of MAA Project NExT Session** *Jan 2024*
Pepperdine University
 - “Making Student Thinking Visible with Team-Based Inquiry Learning”
- MAA MathFest 2022, Co-Organizer of Themed Contributed Paper Session** *Aug 2022*
University of Arizona
 - “Math Circles: Talks about Mathematical Joy, Inspirations, and Data-Driven Lessons Learned”
- Reviewer for:**
 - Journal of Statistics and Data Science Education *Feb 2026*
 - Applied Network Science *Dec 2025*
 - Complexity *March 2023*

- Indian Journal of Discrete Mathematics Nov 2020
- DNA and Cell Biology Jan 2020
- Revista de Matemática: Teoría y Aplicaciones Oct 2019

• **Mathematics and MATLAB Summer Workshop, Co-Coordinator** June 2016 – June 2018
Claremont Graduate University

• **Mathematics and MATLAB Summer Workshop, Co-Instructor** June 2016 – June 2017
Claremont Graduate University

Outreach

• **How I Found My Network: My Path to Mathematics** Nov 2021
Arizona State University
– Keynote address for Sonia Kovalevsky Day

• **Tucson Math Circle** Aug 2019 – Present
University of Arizona
– Co-develop materials and co-run the university sponsored weekly program designed to get middle school students excited about mathematics through hands-on exploration and discovery

• **Association for Women in Mathematics (AWM): Sonia Kovalevsky Day** April 2021
University of Arizona
– Developed materials and co-ran a workshop designed to bolster female high school and middle school students' passion and enthusiasm for mathematics in a supportive environment

• **The Seven Bridges of Königsberg** Nov 2022
Pepperdine University
– Talk given to Math Club during Tuesday Tea

• **Using Network Centrality Measures to Identify Unknown Regulatory Pathways in Pediatric Glioma** Sept 2020
University of Arizona
– Talk given to The MathCats Club (undergraduate math club)

HONORS AND AWARDS

The Teaching and Service Award April 2022
University of Arizona, Department of Mathematics

The Jean E. Miller Excellence in Teaching Award June 2018
The Webb Schools of California

The Thompson and Vivian Webb Excellence in Teaching Award June 2015
The Webb Schools of California

The Heinrich W. Brinkmann Mathematics Prize June 2012
Swarthmore College

FUNDING

Research Grants

• **Academic Year Undergraduate Research Initiative (\$1,000)** Sept 2025, Sept 2023
Pepperdine University

- **Collaborative Research Grant for Postdocs** (\$1,500)
University of Arizona

June 2020

Travel Awards

- **AIM SQuaRE Research Collaboration**
American Institute of Mathematics (AIM)
- **TDA-BIO** (\$1,000)
ACM Conference on Bioinformatics, Computational Biology, and Health Informatics

April 2025

Oct 2016

Fellowships

- **Clinic Fellowship** (\$900)
University of California, Irvine
- **Daniel Pick Fellowship** (\$10,000)
Claremont Graduate University
- **Joseph and Elizabeth Peeler Endowed Fellowship** (\$32,570)
Claremont Graduate University
- **CGU Mathematics Fellowship** (\$13,700)
Claremont Graduate University
- **CGU Minority Fellowship** (\$2,000)
Claremont Graduate University

Jan 2020

Oct 2017

Aug 2015 – June 2017

Aug 2014 – June 2015, June 2017

Aug 2014 – June 2016

SKILLS

Programming Languages

- C (*Moderate proficiency*)
- C++ (*Moderate proficiency*)

- MATLAB (*Proficient*)
- Python (*Proficient*)
- R (*Proficient*)

Scientific Applications

- GitHub

- LaTeX
- RSweave